# On some problems of subcube partitions

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**Abstract**

Let $f(d)$ be the number of partitions of the $d$-dimensional hypercube into subcubes, and let $f_S(d)$ count those whose part dimensions belong to $S$. We determine the second-order asymptotics of $\log f(d)$, proving that $\log f(d) = 2^{d-1}(\log d - 1 + \gamma) + O(2^d d^{-1/4})$, where $\gamma = -\log(\sqrt{3}-1) - 1 + \sqrt{3}/2$. The same estimate holds when every part is an edge or a square. Moreover, allowing subcubes of dimension at least three increases the number of partitions by a factor of only $\exp(o(2^d))$, resolving some problems of Alon, Balogh, and Potapov.

We also determine the typical structure of a uniformly random subcube partition, showing that the proportions of vertices covered by edges and squares converge to $\sqrt{3}-1$ and $2-\sqrt{3}$, respectively. Furthermore, for every fixed $r \geq 2$, we prove that $\log f_r(d) = (r/2^r + o(1))2^d \log d$, where $f_r(d)$ counts partitions into $r$-dimensional subcubes. Our results also yield asymptotic estimates for partitions of $q$-ary hypercubes. The proofs combine entropy methods for hypergraph matchings with weighted enumeration of square packings.

## 1. Introduction

Let $Q_d$ be the *d-dimensional hypercube*, with vertex set $\{0,1\}^d$ and two vertices adjacent whenever they differ in exactly one coordinate. A *subhypercube* (or *subcube* for short) is obtained by fixing some coordinates and leaving the others free. Its *dimension* is the number of free coordinates. A subcube of dimension zero consists of a single vertex, called a *singleton*. A *subcube partition* (or *tiling*) is a partition of $V(Q_d)$ into the vertex sets of subcubes.

One motivation for studying subcube partitions comes from Boolean networks and satisfiability. Bridoux, Gadouleau, and Theyssier [5] established a bijection between subcube partitions and bijective globally commutative Boolean networks, motivating the counting problem subsequently studied by Alon, Balogh, and Potapov [1]. Subcube partitions also correspond to unsatisfiable hitting CNF formulas, which have been studied in connection with satisfiability and proof complexity [14, 19, 23, 27].

Another connection arises in the study of Boolean functions through decision trees. Every decision tree induces a subcube partition, since the inputs reaching each leaf form a subcube. However, not every subcube partition arises in this way. Friedgut, Kahn, and Wigderson [31] studied randomized subcube partitions in connection with graph properties. Kothari, Racicot-Desloges, and Santha [32] showed that subcube partitions can be more efficient than decision trees for computing Boolean functions, and Ambainis, Kokainis, and Kothari [33] established nearly quadratic separations between the corresponding complexity measures. Hegyv'ari [34] studied partitions associated with Boolean functions using additive combinatorial methods.

Several other works concern subcube partitions subject to additional structural restrictions. Filmus et al. [13] investigated extremal properties of tight irreducible subcube partitions.

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Potapov [35] studied perfect clique matchings of $[q]^d$, which correspond to partitions into one-dimensional coordinate subcubes, and determined the order of the logarithm of their number. Tarannikov [36] proved that, for fixed $q$ and $m$, the number of partitions of $[q]^d$ into subcubes of dimension $d-m$ is asymptotic to $d^{(q^m-1)/(q-1)}$. In this setting, every partition has exactly $q^m$ parts, whereas partitions into subcubes of fixed dimension have exponentially many parts.

Related extremal problems concern partitions and coverings of discrete boxes. Alon, Bohman, Holzman, and Kleitman [2] proved that every partition of a $d$-dimensional discrete box into nontrivial sub-boxes has at least $2^d$ parts. Bucić et al. [37] studied partitions of odd-sized discrete boxes into odd-sized proper sub-boxes. In a related covering problem, Kuang and Wang [22] proved that $\{0,1,2\}^d$ can be covered by at most $2(3/2)^d - 1$ binary subcubes.

There are also connections with finite geometry. Identifying $\{0,1\}^d$ with $\mathbb{F}_2^d$, every coordinate subcube is an affine subspace. Bamberg et al. [38] studied partitions into arbitrary affine subspaces, while Potapov, Taranenko, and Tarannikov [30] obtained asymptotic estimates for the number of partitions into two-dimensional affine subspaces in connection with bent functions. D'haeseleer, Ihringer, and Schmidt [10] studied generalized ovoids in polar spaces as a natural $q$-analogue of subcube partitions.

Following Alon, Balogh, and Potapov [1], let $f_S(d)$ count the partitions whose part dimensions belong to $S \subseteq \{0,1,\dots,d\}$, and let $f(d)$ count all subcube partitions. We write $f_{s_1,\dots,s_k}(d)$ for $f_{\{s_1,\dots,s_k\}}(d)$ and $f_{\le r}(d)$ for $f_{\{0,1,\dots,r\}}(d)$. In particular, $m(d) := f_1(d)$ counts perfect matchings of $Q_d$, while $m'(d) := f_{\le 1}(d)$ counts all matchings. Let $\mathcal{P}_S(d)$ be the family counted by $f_S(d)$, and write $\mathcal{P}(d) = \mathcal{P}_{\{0,1,\dots,d\}}(d)$. We set $n = 2^{d-1}$, the size of each bipartition class of $Q_d$, and use the abbreviations $Z_d = f_{\le 2}(d)$ and $R_d = f_{1,2}(d)$. A two-dimensional subcube is a *2-cube* in the terminology of [1]. Throughout this paper, we call it a *square* for convenience.

A natural starting point is the classical enumeration of perfect matchings of the hypercube. Graham and Harary [18] calculated $m(d)$ for small dimensions, and Östergård and Pettersson [26] subsequently enumerated perfect matchings of $Q_7$. The asymptotic scale of $m(d)$ follows from the theory of permanents. The Bregman–Minc inequality [6, 28] shows an upper bound, whereas the van der Waerden permanent theorem [11, 12] and Schrijver's theorem [29] obtain lower bounds. In particular,

$$\log m(d) = n(\log d - 1) + O\left(\frac{n \log d}{d}\right). \tag{1.1}$$

Bounds on matchings in more general regular graphs, including those of Alon and Friedland [3], Friedland and Gurvits [16], and Csikvári [7], form a wider background to this enumeration problem. Further extremal matching estimates were obtained by Davies, Jenssen, and Perkins [8]. Related partition-function questions for independent sets on the hypercube were studied in [17, 20], following the entropy framework of Kahn [21]. The distinction between perfect and nonperfect matchings is smaller. Alon, Balogh, and Potapov [1] proved that $m'(d)/m(d) = \exp(\Theta(n/\sqrt{d}))$, showing that allowing isolated vertices does not change the coefficient of $n$ in (1.1).

The situation changes when squares are allowed. Alon, Balogh, and Potapov [1] proved that, with $N = (d/e)^n$, one has

$$N \exp(\Omega(n)) \le Z_d \le f(d) \le (d+1)^n = N \exp((1+o(1))n). \tag{1.2}$$

They also showed that partitions without squares satisfy $f_{0,1,3,4,\dots,d}(d) \le N \exp(O(nd^{-1/4}))$. This shows that squares generate the additional exponential factor, while singletons and cubes of dimension at least three do not. However, these estimates leave the coefficient of $n$ in $\log f(d)$ undetermined.

### *1.1. Main results*

In this paper, we address the following problems arising from [1]. The first two are stated here at a finer level of precision.

**Problem 1.1** (Alon, Balogh, and Potapov [1])**.**

(i) *Determine* $f(d)$ *and* $f_{\le 2}(d)$ *up to factors of* $\exp(o(n))$ [1, Problem 1.1(iii)–(iv)].

(ii) *Determine the limits* $\lim_{d\to\infty} n^{-1}\log(f_{\le 2}(d)/m'(d))$ *and* $\lim_{d\to\infty} n^{-1}\log(f(d)/m(d))$ [1, Problem 1.2(ii), (iv)].

(iii) *Determine whether* $f(d)/f_{\le 2}(d) = \exp(o(n))$ [1, Problems 1.2(iii) and 1.10].

(iv) *Determine the first-order asymptotics of* $\log f_2(d)$ [1, Problem 1.9].

For (i), Alon, Balogh, and Potapov [1, Propositions 1.5 and 1.6] obtained the bounds in (1.2). These determine the leading term $n\log d$ in both logarithms, but leave the coefficients of $n$ undetermined.

For the ratios in (ii), their results [1, Propositions 1.3–1.6] already imply that both are $\exp(\Theta(n))$. Thus their exponential order is known, while the precise rates remain to be determined.

Problem (iii) concerns the effect of allowing parts of dimension at least three when squares are already allowed. They left (iii) open.

For (iv), Alon, Balogh, and Potapov [1, Proposition 1.7] proved that $f_{0,2}(d) = N^{1+o(1)}$ and expected the same estimate for $f_2(d)$. Their result also treats $f_{0,r}(d)$ for every fixed $r \ge 2$, with a lower bound obtained by counting almost perfect matchings in uniform hypergraphs [4, 15]. The remaining question is whether forbidding singleton parts changes the leading term.

Define

$$p = \sqrt{3} - 1, \qquad \alpha = \frac{1-p}{2} = 1 - \frac{\sqrt{3}}{2}, \qquad \gamma = -\log p - \alpha. \tag{1.3}$$

Here $p^2 = 2(1-p)$ and $\gamma \approx 0.17793$. The constants $p$ and $\alpha$ describe the typical numbers of edges and squares in Theorem 1.4.

**Theorem 1.2.** *There exists an absolute constant* $C > 0$ *such that, for all sufficiently large* $d$*,*

$$\begin{aligned} n(\log d - 1 + \gamma) - C\,n\,d^{-1/3}(\log d)^{4/3} &\le \log R_d \le \log Z_d \le \log f(d) \\ &\le n(\log d - 1 + \gamma) + C\,n\,d^{-1/4}. \end{aligned} \tag{1.4}$$

*Moreover,* $\log Z_d \le n(\log d - 1 + \gamma) + C\,n\,d^{-1/3}$*.*

Theorem 1.2 gives $\log f(d) = n(\log d - 1 + \gamma) + O(nd^{-1/4})$, with the same estimate for $Z_d$, $R_d$, and $f_{1,2,\dots,d}(d)$. Therefore partitions into edges and squares already attain the exponential rate of all subcube partitions, resolving Problem 1.1 (i). Together with (1.1) and $m'(d)/m(d) = \exp(o(n))$, this shows that both limits in Problem 1.1 (ii) equal $\gamma$.

Theorem 1.2 immediately implies the following corollary.

**Corollary 1.3.** *As* $d \to \infty$*,* $\log(f(d)/Z_d) = O(nd^{-1/4})$ *and* $\log(Z_d/R_d) = O(nd^{-1/3}(\log d)^{4/3})$*. Thus,* $f(d)/Z_d = \exp(o(n))$*.*

The first estimate answers Problem 1.1(iii) affirmatively. Allowing subcubes of dimension at least three increases the number of partitions by a factor of $\exp(o(n))$. The second estimate shows that allowing singletons in addition to edges and squares also increases the count by a factor of $\exp(o(n))$. We next describe how the vertices are distributed among parts of different dimensions in a uniformly random partition.

**Theorem 1.4.** *Let* $P_d$ *be uniformly distributed on any one of* $\mathcal{P}_{\{1,2\}}(d)$*,* $\mathcal{P}_{\{0,1,2\}}(d)$*, and* $\mathcal{P}(d)$*. Let* $S_d$ *be its number of square parts,* $E_d$ *its number of edge parts,* $U_d$ *the number of singleton vertices, and* $W_d$ *the number of vertices in parts of dimension at least three. Then* $S_d/n \to \alpha$*,* $E_d/n \to p$*,* $U_d/n \to 0$*, and* $W_d/n \to 0$ *in probability.*

With probability tending to one, a proportion $p + o(1)$ of all vertices lie in edge parts and a proportion $1 - p + o(1)$ lie in square parts. Hence edges and squares together cover all but $o(2^d)$ vertices, even when all part dimensions are allowed. This conclusion allows singleton and higher-dimensional parts to occur; their combined vertex proportion tends to zero.

Our final counting result concerns partitions in which every part has the same fixed dimension $r \geq 2$.

**Theorem 1.5.** *For every fixed integer $r \geq 2$, as $d \to \infty$,*

$$\log f_r(d) = \left( \frac{r}{2^{r-1}} + o(1) \right) n \log d. \tag{1.5}$$

For $r = 2$, Theorem 1.5 gives $\log f_2(d) = (1 + o(1))n \log d$, resolving Problem 1.1 (iv) at first order. For every fixed $r \geq 2$, it shows that forbidding singleton parts preserves the leading term obtained for $f_{0,r}(d)$ in [1, Proposition 1.7]. The error term here is $o(n \log d)$. Therefore the sharp coefficient of $n$ remains open for every fixed $r \geq 2$.

*1.2. Applications*

Our results also give asymptotic estimates and describe the typical structure of bijective globally commutative Boolean networks and unsatisfiable hitting CNF formulas. We further determine the leading asymptotics and typical part dimensions for partitions of $q$-ary hypercubes.

*Boolean networks.* Let $A(d)$ count bijective globally commutative Boolean networks on $d$ labeled coordinates. The correspondence of Bridoux, Gadouleau, and Theyssier [5] identifies each network with a subcube partition. On each part, the network flips precisely the free coordinates. This yields $A(d) = f(d)$. Theorem 1.2 then determines $\log A(d) = n(\log d - 1 + \gamma) + O(nd^{-1/4})$. Theorem 1.4 also describes a uniformly random network $F_d$. The proportions of configurations $x$ with Hamming distance one or two to $F_d(x)$ converge in probability to $p$ and $(1-p)$, respectively. The proportion of fixed configurations tends to zero.

*Hitting formulas.* Let $H(d)$ count unsatisfiable hitting CNFs on a fixed labeled set of $d$ variables, allowing unused variables and regarding clauses as unordered and non-tautological. The subcube correspondence gives $H(d) = f(d)$, so the same logarithmic estimate holds for $H(d)$. In a uniformly random such formula, the number $M_d$ of clauses satisfies $M_d/n \to \sqrt{3}/2$ in probability. The fractions of clauses of widths $(d-1)$ and $(d-2)$ converge to $(2 - 2/\sqrt{3})$ and $(2/\sqrt{3} - 1)$, respectively. Indeed these clauses correspond to edge and square parts, so their limiting fractions are $p/(p+\alpha)$ and $\alpha/(p+\alpha)$; Theorem 1.4 implies that all other parts together contribute $o(n)$ clauses. The fixed-dimension theorem similarly counts unsatisfiable hitting formulas in which every clause has width $(d - r)$. These are enumerative consequences of the correspondence studied in [19, 14, 27].

*$q$-ary hypercubes.* The layer construction also applies to coordinate subcubes of $[q]^d$. Let $f_{\mathcal{D}}^{(q)}(d)$ count partitions with dimensions in a fixed set $\mathcal{D} \subseteq \mathbb{Z}_{\geq 0}$. Potapov [35] showed that the number of perfect clique matchings has logarithmic order $q^d \log d$. For even $q$, he also obtained an upper bound with leading coefficient $1/q$. Combining our layer construction with the matching count of Asratian and Kuzjurin [4] determines the leading coefficient for arbitrary dimension restrictions.

**Corollary 1.6.** *Fix $q \geq 2$ and a set $\mathcal{D} \subseteq \mathbb{Z}_{\geq 0}$ containing a positive integer. Write $\beta = \max_{r \in \mathcal{D}} r/q^r$ and $I = \{r \in \mathcal{D} : r/q^r = \beta\}$. Then $\log f_{\mathcal{D}}^{(q)}(d) = (\beta + o(1))q^d \log d$. In a uniformly random partition, parts with dimensions in $I$ cover all but $o(q^d)$ vertices with probability tending to one.*

For one-dimensional parts, this yields $\log f^{(q)}_{\{1\}}(d) = (1/q + o(1))q^d \log d$, refining the order estimate in [35]. For every $q \geq 3$, one-dimensional parts cover almost all vertices when all dimensions are allowed. In contrast, for $q = 2$ the coefficient $r/2^r$ has two maximizers, $r = 1$ and $r = 2$. The sharper estimate in Theorem 1.2 resolves this tie and determines their individual proportions.

A related consequence concerns partitions with fixed coverage. Let $0 < \theta < 1$ be fixed. The number $K_{q,\theta}(d)$ of matchings formed by $\lfloor \theta q^{d-1} \rfloor$ coordinate lines satisfies $\log K_{q,\theta}(d) = (\theta/q + o(1))q^d \log d$. All proofs appear in Appendix B.

The proof combines a weighted four-uniform encoding with a uniform entropy bound on residual vertex sets. For the lower bound, a deletion recursion estimates square-vacancy probabilities, the permanent theorem supplies many residual matchings, and layer gluing removes the remaining singleton parts.

*Organization.* Section 2 establishes matching-completion and layer-gluing lemmas. Section 3 proves the entropy upper bound, and Section 4 develops the weighted square-packing lower bound. Section 5 treats higher-dimensional parts and completes Theorem 1.2. Section 6 establishes the typical-partition result, and Section 7 considers pure subcube tilings. The quantitative entropy estimate for labeled hypergraph matchings appears in Appendix A, and Appendix B proves the further counting consequences.

## 2. Preliminary

An $r$-uniform hypergraph is a hypergraph in which every edge contains exactly $r$ vertices. We also allow multihypergraphs, where distinct hyperedges may have the same vertex set. For a vertex $v$ of a multihypergraph $\mathcal{H}$, its *degree* is $d_{\mathcal{H}}(v) = |\{e \in E(\mathcal{H}) : v \in e\}|$. For distinct vertices $u, v$, their *codegree* is $d_{\mathcal{H}}(u, v) = |\{e \in E(\mathcal{H}) : u, v \in e\}|$. The maximum degree and maximum codegree are $\Delta(\mathcal{H}) = \max_v d_{\mathcal{H}}(v)$ and $\Delta_2(\mathcal{H}) = \max_{u \neq v} d_{\mathcal{H}}(u, v)$, respectively. For $A \subseteq V(Q_d)$, let $Q_d[A]$ denote the subgraph of $Q_d$ induced by $A$. For $X \subseteq V(Q_d)$, write $Q_d - X = Q_d[V(Q_d) \setminus X]$.

A *matching* is a collection of pairwise vertex-disjoint hyperedges, and it is *perfect* if it covers every vertex. We write $M(\mathcal{H})$ for the number of perfect matchings of $\mathcal{H}$. Throughout, parallel hyperedges are counted with multiplicity, including when counting degrees, codegrees, and perfect matchings.

We use the convention $0 \log 0 = 0$ and write $h(x) = -x \log x - (1-x) \log(1-x)$ for the binary entropy function. Vertices of $Q_d$ are even or odd according to the parity of their coordinate sum. A set $A \subseteq V(Q_d)$ is *balanced* if it contains equally many vertices of each parity. We write $Z_d(A)$ and $R_d(A)$ for the numbers of partitions of $A$ into parts of dimensions in $\{0, 1, 2\}$ and $\{1, 2\}$, respectively. For a balanced set $A$, let $C_d(A, s)$ count the partitions of $A$ into edges and squares containing exactly $s$ squares. An empty set has exactly one partition. Implicit constants are absolute unless a dependence is indicated, and asymptotic statements concern $d \to \infty$.

We establish two auxiliary lemmas used in the lower-bound arguments. The first provides a matching count for bipartite graphs, while the second constructs higher-dimensional subcube partitions from partitions with a common singleton set. For an $n \times n$ matrix $A = (a_{ij})$, its *permanent* is defined by

$$\operatorname{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i,\sigma(i)},$$

where $S_n$ denotes the set of all permutations of $[n]$. We say that $A$ is *doubly stochastic* if $a_{ij} \geq 0$ for all $i, j \in [n]$ and every row and column of $A$ sums to 1.

**Theorem 2.1** (Van der Waerden's permanent theorem [11])**.** *Let $A = (a_{ij})$ be an $n \times n$ doubly stochastic matrix. Then*

$$\operatorname{per}(A) \geq \frac{n!}{n^n},$$

*with equality if and only if $a_{ij} = 1/n$ for all $i, j \in [n]$.*

We begin with a regularization lemma for bipartite graphs. It provides a lower bound on the number of matchings while keeping the number of unmatched vertices small.

**Lemma 2.2.** *Let $H$ be a bipartite graph with $b$ vertices in each part and maximum degree at most $q$, where $q \geq 1$ is an integer. Set $L = bq - |E(H)|$ and $a = \lceil L/q \rceil$. Then $H$ contains at least*

$$\left(\frac{q}{e}\right)^b \exp(-a(1 + \log q)) \tag{2.1}$$

*matchings that leave at most $a$ vertices uncovered in each part.*

*Proof.* Add $a$ vertices to each part of $H$. For each original vertex $v$, add $q - d_H(v)$ edges joining $v$ to the new vertices in the opposite part. Since $\sum_{v \in A}(q - d_H(v)) = L \leq aq$ for either part $A$ of $H$, these edges can be chosen so that every new vertex has degree at most $q$. We then add edges between the new vertices to obtain a $q$-regular bipartite multigraph $\widehat{H}$.

Let $A$ be the adjacency matrix of $\widehat{H}$, where each entry counts the number of edges between the corresponding vertices. Since $\widehat{H}$ is $q$-regular, $A/q$ is doubly stochastic. By Theorem 2.1, the number of perfect matchings in $\widehat{H}$, counted with edge multiplicity, is

$$\operatorname{per}(A) = q^{b+a} \operatorname{per}(A/q) \geq q^{b+a} \frac{(b+a)!}{(b+a)^{b+a}} \geq \left(\frac{q}{e}\right)^{b+a}.$$

Let $P$ be a perfect matching of $\widehat{H}$, and let $M = P \cap E(H)$. Every vertex of $H$ not covered by $M$ is matched in $P$ to a new vertex in the opposite part. Since each part contains only $a$ new vertices, $M$ leaves at most $a$ vertices uncovered in each part. Conversely, fix a matching $M$ of $H$. Any perfect matching $P$ of $\widehat{H}$ satisfying $P \cap E(H) = M$ is determined by the edges of $P$ incident with the $2a$ new vertices. Since each new vertex has degree $q$, there are at most $q^{2a}$ such perfect matchings. Therefore, the number of matchings in $H$ leaving at most $a$ vertices uncovered in each part is at least

$$\frac{(q/e)^{b+a}}{q^{2a}} = \left(\frac{q}{e}\right)^b \exp(-a(1 + \log q)).$$

□

A related coordinate-extension construction appears in Potapov [35] for perfect clique matchings. The following lemma gives a lower bound for the number of subcube partitions by combining partitions with the same singleton set.

**Lemma 2.3.** *Let $k, r$ be positive integers and $\mathcal{D} \subseteq [k]$. Let $\mathcal{A} \subseteq \mathcal{P}_{\{0\} \cup \mathcal{D}}(k)$ be a nonempty family of subcube partitions of $Q_k$. For each $S \subseteq V(Q_k)$, let $a(S)$ be the number of partitions in $\mathcal{A}$ whose singleton parts are exactly $\{\{v\} : v \in S\}$, and let $K = |\{S : a(S) > 0\}|$. Then*

$$f_{\mathcal{D} \cup \{r\}}(k + r) \geq \sum_S a(S)^{2^r} \geq \frac{|\mathcal{A}|^{2^r}}{K^{2^r - 1}}. \tag{2.2}$$

*Proof.* Identify $Q_{k+r}$ with $Q_k \times Q_r$. Fix $S \subseteq V(Q_k)$ with $a(S) > 0$. For each $y \in Q_r$, choose a partition $\mathcal{P}_y \in \mathcal{A}$ whose singleton parts are exactly $\{\{v\} : v \in S\}$. We form a partition of $Q_{k+r}$ by taking $C \times \{y\}$ for every positive-dimensional part $C \in \mathcal{P}_y$ and $\{v\} \times Q_r$ for every $v \in S$.

All parts have dimensions in $\mathcal{D} \cup \{r\}$. Moreover, the intersection of the resulting partition with each $Q_k \times \{y\}$ recovers $\mathcal{P}_y$. Thus distinct choices of the partitions $\mathcal{P}_y$ give distinct partitions of $Q_{k+r}$. Summing over $S$ proves the first inequality in (2.2).

Since $\sum_S a(S) = |\mathcal{A}|$, Hölder's inequality gives $|\mathcal{A}|^{2^r} \leq K^{2^r - 1} \sum_S a(S)^{2^r}$, which proves the second inequality. □

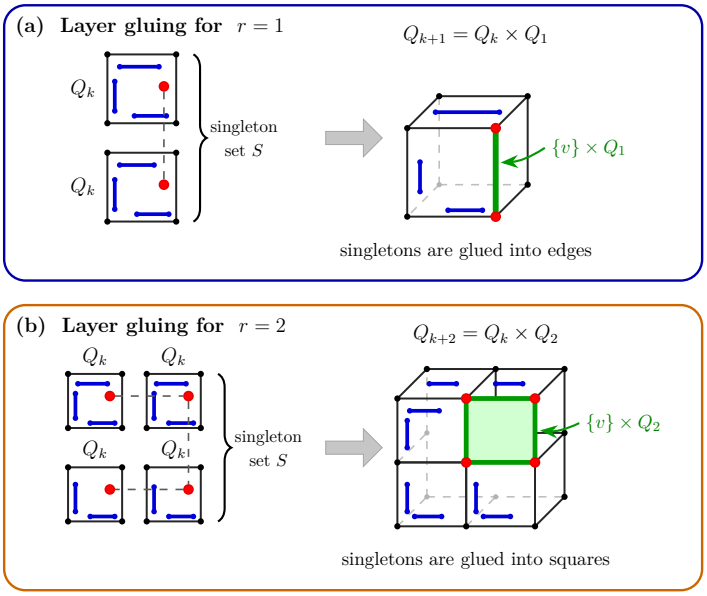



Figure 1: Illustration of proof for subcube partitions (Lemma 2.3). (a) For $r = 1$, parallel layers $Q_k$ sharing a common singleton set $S$ glue corresponding singleton vertices into edge parts $\{v\} \times Q_1$. (b) For $r = 2$, four parallel layers sharing $S$ glue corresponding singletons into square parts $\{v\} \times Q_2$, eliminating uncovered defects.

## 3. Uniform upper bounds on edge and square partitions

### *3.1. Directional encodings*

We use the encoding of Alon, Balogh, and Potapov [1, proof of Proposition 1.6] in the following form.

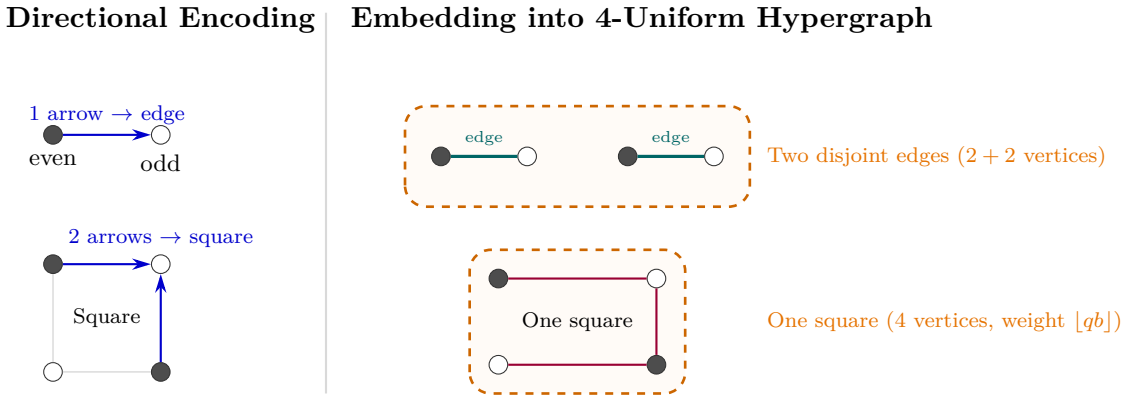



Figure 2: The two constructions in Section 3. Left shows the directional encoding where arrows identify edges and squares. Right shows how edges and squares become uniform blocks of size four.

**Lemma 3.1.** *Let $A \subseteq V(Q_d)$ be a balanced subset with $b$ vertices of each parity. Then $R_d(A) \leq d^b$. Furthermore, if $Z_{d,t}$ counts partitions in $\mathcal{P}_{\{0,1,2\}}(d)$ with $t$ singletons of each parity, then*

$$Z_{d,t} \leq \binom{n}{t} d^{n-t}. \tag{3.1}$$

*Proof.* Let $v_1, \ldots, v_b$ be the even vertices of $A$ in a fixed order. For each partition $\mathcal{P}$ counted by $R_d(A)$, we associate a sequence $(s_1, \ldots, s_b) \in [d]^b$ as follows. If $v_i$ belongs to a part $C$, let $u$ be the lexicographically first odd vertex of $C$, and let $s_i$ be the coordinate in which $v_i$ and $u$ differ. This is well-defined since every part has dimension one or two.

The sequence uniquely determines $\mathcal{P}$. Indeed, each $s_i$ specifies an odd neighbor of $v_i$. An odd vertex specified once determines an edge, while an odd vertex specified twice determines a square, with its two coordinate directions given by the corresponding entries of the sequence. Thus $R_d(A) \le d^b$.

For (3.1), first choose the $t$ even vertices that form singleton parts. The remaining $n-t$ even vertices can be encoded by a sequence over $[d]$ as above. This sequence determines all parts of positive dimension, and every remaining vertex forms a singleton part. Hence $Z_{d,t} \le \binom{n}{t} d^{n-t}$. □

*3.2. Entropy bounds from hypergraph matchings*

We estimate the number of edge–square partitions with a prescribed number of squares. For $q > 0$ and $0 \le x \le 1/2$, define

$$\Psi_q(x) = \frac{1}{2}\log(1+q/2) - x - \frac{1-2x}{2}\log(1-2x) - x\log q. \tag{3.2}$$

Here $x$ represents the number of squares divided by the size of each parity class, while $q$ is an auxiliary weighting parameter. The following estimate identifies $\Psi_q(x)$ as the correction to the leading exponent.

The following proposition bounds the number of edge–square partitions with a prescribed number of squares, uniformly over balanced subsets of $Q_d$.

**Proposition 3.2.** *Fix $q > 0$. For all sufficiently large $d$, every balanced set $A \subseteq V(Q_d)$ with $b \ge 2d$ vertices of each parity and every integer $0 \le s \le \lfloor b/2 \rfloor$ satisfy*

$$C_d(A,s) \le \exp\left(b\left(\log d - 1 + \Psi_q(s/b)\right) + O_q(bd^{-1/3} + \log b)\right). \tag{3.3}$$

*The error term is uniform over $A$ and $s$. In particular,*

$$C_d(A,s) \le \exp\left(b(\log d - 1 + \gamma) - \frac{(s-\alpha b)^2}{b} + O(bd^{-1/3} + \log b)\right), \tag{3.4}$$

*uniformly over the same choices of $A$ and $s$.*

*Proof.* We begin with the case of even $b$. Define a labeled 4-uniform multihypergraph $\mathcal{H}$ on $A$ as follows. For each unordered pair of disjoint edges of $Q_d[A]$, we include a hyperedge labeled by these two edges. For each square contained in $A$, include $\lfloor qb \rfloor$ distinguishable copies of its vertex set, which is positive for large $d$. Hyperedges with the same vertex set but different labels are counted separately.

The graph $Q_d[A]$ contains at most $bd$ edges and has maximum degree at most $d$. Thus, each vertex belongs to at most $K = bd^2(1+\frac{q}{2})$ labeled hyperedges. Now take two distinct vertices in $A$. Pairs of edges contain both vertices in at most $(bd + d^2)$ ways, while squares contain both vertices in at most $\lfloor qb \rfloor (d-1)$ ways. Since $b \ge d$, the maximum codegree satisfies $\Delta_2(\mathcal{H}) = O_q(bd)$.

By the entropy bound in Lemma A.1 with hyperedge size four and ratio $\delta = O_q(d^{-1})$, the number $M(\mathcal{H})$ of labeled perfect matchings satisfies

$$M(\mathcal{H}) \le \left(\frac{bd^2(1+\frac{q}{2})}{e^3}\right)^{b/2} \exp(O_q(bd^{-1/3})). \tag{3.5}$$

Now consider any partition of $A$ containing $s$ squares and $b-2s$ edges. We can lift this partition to labeled perfect matchings of $\mathcal{H}$. To do so, group the $b-2s$ edges into disjoint pairs in $(b-2s-1)!!$ ways, where $(-1)!! = 1$. Next, assign one of the $\lfloor qb \rfloor$ labels to each of the $s$ squares, giving $\lfloor qb \rfloor^s$ choices. Summing over all possible square counts yields

$$M(\mathcal{H}) = \sum_{j=0}^{b/2} C_d(A,j)(b-2j-1)!!\,\lfloor qb \rfloor^j. \tag{3.6}$$

Each labeled perfect matching determines the partition, the edge pairs, and the square labels uniquely. Write $u = 1 - \frac{2s}{b}$. By the Stirling approximation, uniformly for all $s$ including $b - 2s = 0$,

$$\log((b-2s-1)!!\,\lfloor qb \rfloor^s) = \frac{b}{2}\log b + \frac{bu}{2}\log u - \frac{bu}{2} + s\log q + O_q(\log b). \tag{3.7}$$

Rounding $\lfloor qb \rfloor$ to an integer changes this quantity by at most $O_q(1)$. Dividing the bound on $M(\mathcal{H})$ by the term for index $s$ gives the result for even $b$.

Next assume that $b \geq 2d$ is odd. Because $(b - 2s)$ is odd, every partition in this family must contain at least one edge. Deleting an edge gives

$$C_d(A, s) \leq \sum_{e \in E(Q_d[A])} C_d(A \setminus e, s). \tag{3.8}$$

Each set $A \setminus e$ has $(b-1) \geq d$ vertices of each parity, which is even. The derivative of $u \log u$ is unbounded at zero. However, for $u, v \in [0, 1]$ with $|u - v| \leq 1/b$, we have $|u \log u - v \log v| = O((\log b)/b)$. Since the arguments differ by at most $1/b$, this gives a change of $O_q((\log b)/b)$ in $\Psi_q$. Replacing $b - 1$ with $b$ in the exponent changes the value by $O_q(\log b + \log d)$. Since there are at most $bd$ edges in the sum, taking the logarithm adds another $O(\log b)$ error term, which proves (3.3) for odd $b$. Taking $q = p$, we have $\Psi_p(\alpha) = \gamma$, $\Psi_p'(\alpha) = 0$, and $\Psi_p''(x) = -2/(1-2x) \leq -2$ for $0 \leq x < 1/2$. Consequently, $\Psi_p(x) \leq \gamma - (x-\alpha)^2$ for $0 \leq x \leq 1/2$. Substituting this into (3.3) gives (3.4). □

**Proposition 3.3.** *For every subset $A \subseteq V(Q_d)$ with $m = |A|/2$,*

$$Z_d(A) \leq \exp\left(m(\log d - 1 + \gamma) + O(nd^{-1/3})\right), \tag{3.9}$$

*where the error term is uniform over all choices of $A$.*

*Proof.* Let $A'$ be a balanced subset with $b$ vertices of each parity. If $b < 2d$, Lemma 3.1 gives $R_d(A') \leq d^b$. If $b \geq 2d$, summing the estimate from Proposition 3.2 over all $s$ gives the same upper bound. Therefore, $R_d(A') \leq \exp\left(b(\log d - 1 + \gamma) + O(nd^{-1/3})\right)$, using $b \leq n$ and $d + \log n = O(nd^{-1/3})$. This also holds when $R_d(A') = 0$.

Now let $A$ be an arbitrary subset. Deleting all singletons from a partition counted by $Z_d(A)$ leaves a balanced subset $A'$ of size $2b$. There are at most $\binom{2m}{2m-2b}$ ways to choose the deleted vertices. Writing $a_d = de^{\gamma - 1}$, we apply the bound on $R_d(A')$ to obtain

$$Z_d(A) \leq \exp(O(nd^{-1/3})) \sum_{0 \leq b \leq \lfloor m \rfloor} \binom{2m}{2m-2b} a_d^b \leq \exp(O(nd^{-1/3}))\, a_d^m (1 + a_d^{-1/2})^{2m}. \tag{3.10}$$

Here $2m$ is an integer even when $m$ is not, and the last bound includes the missing parities in the binomial expansion. Because $2m \log(1 + a_d^{-1/2}) = O(nd^{-1/2})$, this finishes the proof. □

## 4. The lower bound via weighted square packings

A *square packing* is a collection $B$ of pairwise vertex-disjoint squares of $Q_d$. Write $V(B)$ for the union of their vertex sets. The *residual graph* of $B$ is the induced subgraph $Q_d - V(B)$.

The deletion recursion used below is related to the correlation-decay approach to counting independent sets and matchings [41, 42, 43]. We use it here to obtain uniform asymptotic vacancy estimates for square packings.

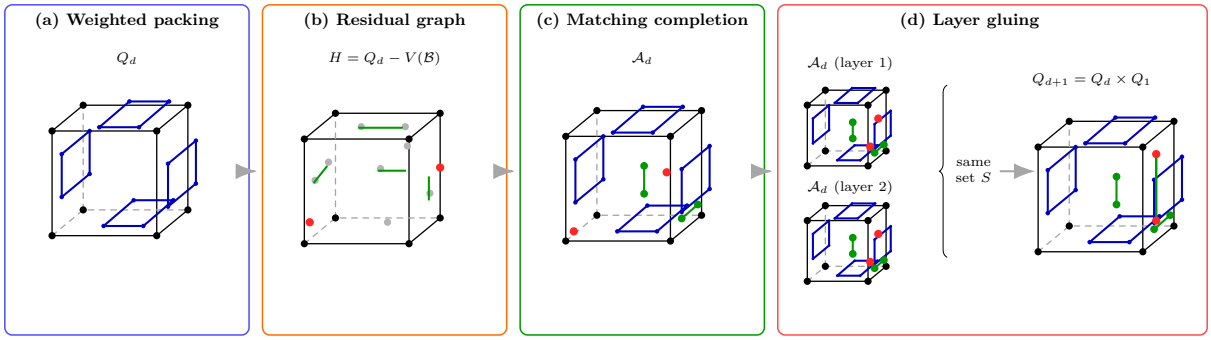



Figure 3: (a) Weighted square packing of typical size $|\mathcal{B}| \approx \alpha n$, with weight proportional to $(c/D)^{|\mathcal{B}|}$. (b) Residual graph $H = Q_d - V(\mathcal{B})$ after removing the packing; most degrees are close to $pd$, with only few exceptional vertices. (c) Matching completion produces many edge-square almost-partitions with few singletons, counted via the permanent bound. (d) Layer gluing identifies layers with the same singleton set $S$. Singletons group into edges, yielding a partition counted by $R_{d+1}$. Together, square packings, matching completion, and layer gluing give the lower bound for $R_d$.

### *4.1. Uncovered probabilities for square packings*

Write $D = \binom{d}{2}$, the number of squares containing any fixed vertex of $Q_d$. For $0 \le t \le c$, where $c = \frac{1-p}{p^4} = \frac{2+\sqrt{3}}{4}$, define the *partition function*

$$I_d(t) = \sum_B \left(\frac{t}{D}\right)^{|B|},$$

where the sum ranges over all collections $B$ of pairwise vertex-disjoint squares of $Q_d$.

Normalizing these weights defines a probability measure $\mathbb{P}_t(B) = (t/D)^{|B|}/I_d(t)$ on square packings, concentrated on the empty packing when $t = 0$. For $U \subseteq V(Q_d)$, let $I_{d,U}(t)$ be the total weight of square packings that cover no vertex of $U$. For $v \in V(Q_d) \setminus U$, define

$$P_{t,U}(v) = \frac{I_{d,U\cup\{v\}}(t)}{I_{d,U}(t)}.$$

Thus $P_{t,U}(v)$ is the probability that $v$ is uncovered, conditional on every vertex of $U$ being uncovered.

**Lemma 4.1.** *For every square $F$ containing $v$, order $F \setminus \{v\}$ as $w_1, w_2, w_3$, and set $W_0 = \varnothing$, $W_1 = \{w_1\}$, $W_2 = \{w_1, w_2\}$. Then*

$$P_{t,U}(v)^{-1} = 1 + \frac{t}{D} \sum_{\substack{F \ni v \\ F\cap U = \varnothing}} \prod_{j=1}^{3} P_{t,U\cup\{v\}\cup W_{j-1}}(w_j). \tag{4.1}$$

*Proof.* It is easy to show that

$$I_{d,U}(t) = I_{d,U\cup\{v\}}(t) + \frac{t}{D} \sum_{\substack{v \in F \\ F\cap U = \varnothing}} I_{d,U\cup F}(t).$$

For a square $F = \{v, w_1, w_2, w_3\}$ in the sum, we have

$$\frac{I_{d,U\cup F}(t)}{I_{d,U\cup\{v\}}(t)} = \frac{I_{d,U\cup\{v,w_1\}}(t)}{I_{d,U\cup\{v\}}(t)} \frac{I_{d,U\cup\{v,w_1,w_2\}}(t)}{I_{d,U\cup\{v,w_1\}}(t)} \frac{I_{d,U\cup F}(t)}{I_{d,U\cup\{v,w_1,w_2\}}(t)} = \prod_{j=1}^{3} P_{t,U\cup\{v\}\cup W_{j-1}}(w_j).$$

Dividing the preceding expression for $I_{d,U}(t)$ by $I_{d,U\cup\{v\}}(t)$ and using $P_{t,U}(v) = I_{d,U\cup\{v\}}(t)/I_{d,U}(t)$ proves (4.1). □

The recursion in Lemma 4.1 suggests a limiting value for the conditional vacancy probabilities. Indeed, when $|U|$ is bounded, the number of squares containing $v$ and avoiding $U$ is $D - O(d)$. If the conditional vacancy probabilities in (4.1) are approximated by a common value $x$, then

$$x^{-1} \approx 1 + \frac{t}{D} \cdot Dx^3 = 1 + tx^3.$$

This motivates the fixed-point equation $x + tx^4 = 1$. For each $t \in [0, c]$, the function $x \mapsto x + tx^4$ is strictly increasing on $[0, 1]$, with values 0 and $1 + t$ at the endpoints. Hence there is a unique $p(t) \in (0, 1]$ satisfying

$$p(t) + tp(t)^4 = 1.$$

In particular, $p(0) = 1$, and $c = (1 - p)/p^4$ implies $p(c) = p$.

**Lemma 4.2.** *Uniformly for all $t \in [0, c]$, all sets $U$ with $|U| \le 3$, and all vertices $v \notin U$, we have*

$$P_{t,U}(v) = p(t) + O(\log d/d). \tag{4.2}$$

*Thus for every subset $S \subseteq V(Q_d)$ with $1 \le |S| \le 4$,*

$$\mathbb{P}_t(\textit{all vertices in } S \textit{ are uncovered}) = p(t)^{|S|} + O(\log d/d). \tag{4.3}$$

*Proof.* Fix $t \in [0, c]$ and let $\ell = \lceil 10 \log d \rceil$ and $\delta_d = (3\ell + 3)(d - 1)/D$. We may assume that $d$ is sufficiently large that $\delta_d < 1$. Iterate the recursion in (4.1) to depth $\ell$. At each step the deleted set increases by at most three vertices. Every deleted set arising in this expansion therefore has size at most $3\ell + 3$. Consequently the current vertex belongs to between $(1 - \delta_d)D$ and $D$ squares disjoint from the deleted set at each step of the recursion.

For $s \ge 0$ define $T_s(x) = (1 + sx^3)^{-1}$. Initialize $a_0 = 0$ and $b_0 = 1$. For $i \ge 0$ define the recursive sequences

$$a_{i+1} = T_t(b_i) \quad \text{and} \quad b_{i+1} = T_{t(1-\delta_d)}(a_i).$$

We claim that $a_\ell \le P_{t,U}(v) \le b_\ell$. Starting with the bounds $0 \le P_{t,W}(x) \le 1$ at the leaves we apply induction towards the root. If each factor on the right hand side of (4.1) lies in $[a_i, b_i]$, then each product lies in $[a_i^3, b_i^3]$. The bounds on the number of summands imply that the corresponding probability lies in $[a_{i+1}, b_{i+1}]$. Because $T_t$ is decreasing and $T_t(p(t)) = p(t)$, induction also yields $a_i \le p(t) \le b_i$ for every $i$.

We next bound the difference $b_i - a_i$. For $s \in [0, c]$ and $x \in [0, 1]$ differentiation yields

$$|T_s'(x)| = \frac{3sx^2}{(1 + sx^3)^2} \le \left(\frac{16c}{27}\right)^{1/3}. \tag{4.4}$$

Let $\lambda$ denote this upper bound, which is strictly less than 1. We also have $0 \le T_{t(1-\delta_d)}(x) - T_t(x) \le c\delta_d$ throughout $[0, 1]$. The mean value theorem therefore implies that $b_{i+1} - a_{i+1} \le \lambda(b_i - a_i) + c\delta_d$. Iterating this inequality yields $b_\ell - a_\ell \le \lambda^\ell + c\delta_d/(1 - \lambda)$. Because $\delta_d = O(\log d/d)$ and $\lambda^\ell = O(d^{-1})$, this difference is $O(\log d/d)$. Since both $P_{t,U}(v)$ and $p(t)$ lie in $[a_\ell, b_\ell]$, the first assertion follows.

Finally write $S = \{v_1, \ldots, v_r\}$ with $r \le 4$. By successive conditioning the probability that every vertex of $S$ is uncovered equals $\prod_{j=1}^{r} P_{t,\{v_1,\ldots,v_{j-1}\}}(v_j)$. Each conditioning set has size at most three. This guarantees that the first assertion applies to each factor, which establishes (4.3). □

*4.2. The partition function*

We determine the partition function by integrating its logarithmic derivative, following the standard occupancy fraction approach; see, for example, [9].

**Lemma 4.3.** *The independence polynomial satisfies*

$$\log I_d(c) = nF + O\left(n\frac{\log d}{d}\right), \tag{4.5}$$

*where* $F = -2\log p - \frac{3}{2}(1-p)$.

*Proof.* Differentiating the partition function and removing a distinguished square gives $I_d'(t) = D^{-1}\sum_C I_{d,C}(t)$, where the sum ranges over all squares $C$ of $Q_d$. There are $nD/2$ such squares. By Lemma 4.2,

$$\frac{I_d'(t)}{nI_d(t)} = \frac{1}{nD}\sum_C \mathbb{P}_t(\text{all vertices of } C \text{ are uncovered}) = \frac{1}{2}p(t)^4 + O\left(\frac{\log d}{d}\right),$$

uniformly for $t \in [0, c]$. Since $I_d(0) = 1$, integration yields

$$\frac{1}{n}\log I_d(c) = \frac{1}{2}\int_0^c p(t)^4\,dt + O\left(\frac{\log d}{d}\right). \tag{4.6}$$

With the substitution $z = p(t)$, we have $t = (1-z)/z^4$, and hence $\frac{1}{2}\int_0^c p(t)^4\,dt = \int_p^1 (2/z - 3/2)\,dz = -2\log p - \frac{3}{2}(1-p) = F$. This proves (4.5). □

*4.3. Matchings in residual graph*

Let $\rho_d = (\log d/d)^{1/3}$. The following lemma shows that a random square packing typically has size close to $\alpha n$ and leaves a bipartite graph in which almost all vertices have degree close to $pd$.

Throughout this subsection, square packings are sampled according to $\mathbb{P}_c$.

Alon, Balogh, and Potapov [1, proof of Proposition 1.5] obtained a lower bound by combining disjoint squares with matchings in the remaining graph. We refine this approach using the weighted square packing model to determine the sharp exponential constant.

**Lemma 4.4.** *There exists an absolute constant $C_0 > 0$ such that, for all sufficiently large $d$, a square packing $B$ sampled according to $\mathbb{P}_c$ satisfies the following properties with probability at least $1/2$. Writing $H = Q_d - V(B)$, we have*

$$\big||B| - \alpha n\big| \le n\rho_d,$$

*and all but at most $C_0 n\rho_d$ vertices $v \in V(H)$ satisfy*

$$|\deg_H(v) - pd| \le d\rho_d.$$

*Proof.* For each $v \in V(Q_d)$, let $I_v$ be the indicator that $v$ is uncovered by $B$. By Lemma 4.2, for distinct vertices $u, v, w$, we have $\mathbb{E}I_v = p + O(\log d/d)$, $\mathbb{E}I_uI_v = p^2 + O(\log d/d)$ and $\mathbb{E}I_uI_vI_w = p^3 + O(\log d/d)$. Since $4|B| = 2n - \sum_v I_v$ and $\alpha = (1-p)/2$, we have $\mathbb{E}|B| = \alpha n + O(n\log d/d)$ and $\mathrm{Var}(|B|) = O(n + n^2\log d/d)$. By Chebyshev's inequality,

$$\mathbb{P}_c\big(\big||B| - \alpha n\big| > n\rho_d\big) \le \frac{4\,\mathrm{Var}(|B|)}{n^2\rho_d^2} = O\left(\frac{1}{n\rho_d^2} + \frac{\log d}{d\rho_d^2}\right) = O(\rho_d) < \frac{1}{4}.$$

Next, fix $v \in V(Q_d)$ and let $Y_v = \sum_{u\sim v} I_u$. Conditional on $I_v = 1$, $d_{Q_d - V(B)}(v) = Y_v$. Thus,

$$\mathbb{E}(Y_v \mid I_v = 1) = \sum_{u\sim v}\frac{\mathbb{E}I_uI_v}{\mathbb{E}I_v} = pd + O(\log d),$$
$$\mathbb{E}(Y_v^2 \mid I_v = 1) = \sum_{u\sim v}\frac{\mathbb{E}I_uI_v}{\mathbb{E}I_v} + \sum_{\substack{u,w\sim v\\ u\ne w}}\frac{\mathbb{E}I_uI_vI_w}{\mathbb{E}I_v} = dp + d(d-1)p^2 + O(d\log d).$$

Hence, $\mathrm{Var}(Y_v \mid I_v = 1) = O(d \log d)$. Since $\log d = o(d\rho_d)$, by Chebyshev's inequality,

$$\mathbb{P}_c(|Y_v - pd| > d\rho_d \mid I_v = 1) = O\left(\frac{\log d}{d\rho_d^2}\right) = O(\rho_d).$$

Let $N_{\mathrm{bad}} = \sum_{v \in V(Q_d)} I_v \mathbf{1}_{\{|Y_v - pd| > d\rho_d\}}$. Then $\mathbb{E}N_{\mathrm{bad}} = \sum_{v \in V(Q_d)} \mathbb{P}_c(I_v = 1,\ |Y_v - pd| > d\rho_d) = O(n\rho_d)$. Choosing $C_0$ sufficiently large, by Markov's inequality,

$$\mathbb{P}_c(N_{\mathrm{bad}} > C_0 n\rho_d) \le \frac{\mathbb{E}N_{\mathrm{bad}}}{C_0 n\rho_d} \le \frac{1}{4}.$$

Hence both conditions hold simultaneously with probability at least $1/2$. □

We now combine the square packings from Lemma 4.4 with matchings in their residual graphs to construct a large family of subcube partitions with few singleton parts.

**Proposition 4.5.** *There exist absolute constants $C_1, C_2 > 0$ such that, for all sufficiently large $d$, there is a family $\mathcal{A}_d \subseteq \mathcal{P}_{\{0,1,2\}}(d)$ in which every partition has at most $C_1 n\rho_d$ singleton vertices of each parity and*

$$\log |\mathcal{A}_d| \ge n(\log d - 1 + \gamma) - C_2 n\rho_d \log d. \tag{4.7}$$

*Proof.* Let $\mathcal{G}_d$ be the family of square packings satisfying the two conditions in Lemma 4.4. Since $\mathbb{P}_c(\mathcal{G}_d) \ge 1/2$, we have $\sum_{B \in \mathcal{G}_d} \left(\frac{c}{D}\right)^{|B|} \ge \frac{1}{2} I_d(c)$. Every $B \in \mathcal{G}_d$ satisfies $|B| \ge \alpha n - n\rho_d$. Since $\frac{c}{D} < 1$ for sufficiently large $d$, $|\mathcal{G}_d| \ge \frac{1}{2} I_d(c) \left(\frac{D}{c}\right)^{\alpha n - n\rho_d}$. By Lemma 4.3,

$$\log |\mathcal{G}_d| \ge nF + \alpha n \log(D/c) - O(n\rho_d \log d). \tag{4.8}$$

Fix $B \in \mathcal{G}_d$ and let $H = Q_d - V(B)$. Since every square contains two vertices of each parity, $H$ has $b = n - 2|B| = pn + O(n\rho_d)$ vertices in each part. Moreover, all but at most $C_0 n\rho_d$ vertices of $H$ have degree in the interval $[pd - d\rho_d, pd + d\rho_d]$.

Let $q_d = \lceil pd + 2d\rho_d \rceil$. By Lemma 4.4, all but at most $C_0 n\rho_d$ vertices of $H$ have degree at most $pd + d\rho_d < q_d$. We delete edges incident with the remaining vertices until every vertex has degree at most $q_d$, obtaining a subgraph $H' \subseteq H$. Since $\Delta(H) \le d$, at most $C_0 nd\rho_d$ edges are deleted.

Set $L = bq_d - |E(H')|$. Summing the degrees over the even vertices of $H$, we obtain $0 \le L = \sum_{\substack{v \in V(H) \\ v \text{ even}}} (q_d - \deg_H(v)) + |E(H) \setminus E(H')| \le b(3d\rho_d + 1) + C_0 n\rho_d q_d + C_0 nd\rho_d = O(nd\rho_d)$.

Since $q_d = pd + O(d\rho_d) = \Theta(d)$ and $L = O(nd\rho_d)$, we may set $a = \lceil L/q_d \rceil = O(n\rho_d)$. By Lemma 2.2, the graph $H'$ has at least $\left(\frac{q_d}{e}\right)^b \exp\left(-\left\lceil \frac{L}{q_d} \right\rceil (1 + \log q_d)\right) \ge \exp(b(\log(pd) - 1) - O(n\rho_d \log d))$ matchings leaving at most $a$ vertices uncovered in each part.

Each such matching, together with the squares of $B$ and the uncovered vertices as singleton parts, determines a partition in $\mathcal{P}_{\{0,1,2\}}(d)$. Hence there is a family $\mathcal{A}_d$ in which every partition has at most $C_1 n\rho_d$ singleton vertices of each parity and $\log |\mathcal{A}_d| \ge \log |\mathcal{G}_d| + pn(\log(pd) - 1) - O(n\rho_d \log d) \ge nF + \alpha n \log(D/c) + pn(\log(pd) - 1) - O(n\rho_d \log d)$. Recall that $D = \binom{d}{2}$ and $2\alpha + p = 1$ and $2c = p^{-2}$. Combining these with $F = -2\log p - \frac{3}{2}(1 - p)$ simplifies the constant terms to $F - \alpha \log(2c) + p \log p - p = -\log p - \alpha - 1 = \gamma - 1$. Substituting this back gives $\log |\mathcal{A}_d| \ge n(\log d - 1 + \gamma) - O(n\rho_d \log d)$, as required. □

*4.4. The lower bound for $R_d$*

We now use the product construction in Lemma 2.3 to obtain partitions without singleton parts.

*Proof.* Let $\mathcal{A} = \mathcal{A}_{d-1}$ be the family from Proposition 4.5. Each parity class of $Q_{d-1}$ has $n/2$ vertices. Since every edge and square contains equally many vertices of each parity, each partition in $\mathcal{A}$ has the same number of even and odd singleton vertices.

Let $K$ be the number of distinct singleton sets among the partitions in $\mathcal{A}$. By Proposition 4.5, each such set contains at most $C_1 n\rho_{(d-1)}/2$ vertices of each parity. Hence $K \le \sum_{0\le t\le C_1 n\rho_{(d-1)}/2} \binom{n/2}{t}^2 \le \exp(O(n\rho_{(d-1)}\log d))$.

By Lemma 2.3, with $r = 1$ and $\mathcal{D} = \{1, 2\}$, $R_d \ge \dfrac{|\mathcal{A}|^2}{K}$. Together with Proposition 4.5, this yields $\log R_d \ge 2\log|\mathcal{A}| - \log K \ge n(\log(d-1) - 1 + \gamma) - O(n\rho_{(d-1)}\log d)$. Since $\log(d-1) = \log d + O(1/d)$ and $\rho_{(d-1)}\log d = O\big(d^{-1/3}(\log d)^{4/3}\big)$, the result follows. □

## 5. Higher-dimensional subcube

In this section, we show that allowing subcubes of dimension at least three increases the number of partitions by at most a factor of $\exp(o(n))$. The proof combines the coordinate encoding of Alon, Balogh, and Potapov [1, proof of Proposition 1.8] with the uniform upper bound for partitions into singletons, edges, and squares on arbitrary vertex subsets established in Section 3.

For $0 \le T \le n$, let $f(d;T)$ count partitions in which the parts of dimension at least three cover $T$ even and $T$ odd vertices. Thus $f(d) = \sum_T f(d;T)$ and $f(d;0) = Z_d$.

**Lemma 5.1.** *Uniformly for* $0 \le T \le n$,

$$f(d;T) \le \exp\big(n(\log d - 1 + \gamma) + O(nd^{-1/3})\big)\binom{n}{T}\big(2e^{1-\gamma}d^{-1/4}\big)^T. \tag{5.1}$$

*Proof.* Consider a partition counted by $f(d;T)$, and let $A_e$ be the set of even vertices covered by its parts of dimension at least three. Since $|A_e| = T$, there are $\binom{n}{T}$ choices for $A_e$.

Fix $A_e$. We use the coordinate encoding of Alon, Balogh, and Potapov [1, proof of Proposition 1.8]. For each part $C$ of dimension $k \ge 3$, let $u_C$ be its lexicographically smallest odd vertex. At each even vertex $v \in C$, record the coordinate $j$ if $v = u_C \oplus e_j$, and record 0 otherwise, where $e_j$ is the $j$th unit vector. Exactly $k$ of the $2^{k-1}$ even vertices of $C$ receive nonzero records. Since $k/2^{k-1} \le 3/4$ for $k \ge 3$, at most $3T/4$ records are nonzero.

These records uniquely determine all parts of dimension at least three. Indeed, each nonzero record $(v, j)$ identifies $u_C = v \oplus e_j$, where $e_j$ is the $j$th unit vector and $\oplus$ denotes coordinatewise addition modulo 2. The coordinate labels corresponding to the same $u_C$ give precisely the free coordinates of $C$. Thus, for a fixed $A_e$, the number of possible collections of such parts is at most

$$\sum_{j=0}^{\lfloor 3T/4\rfloor}\binom{T}{j}d^j \le 2^T d^{3T/4}.$$

Each part of positive dimension contains equally many vertices of each parity. Hence, after removing all parts of dimension at least three, exactly $2(n-T)$ vertices remain, and their parts have dimensions at most two. By Proposition 3.3, these remaining vertices admit at most $\exp((n-T)(\log d - 1 + \gamma) + O(nd^{-1/3}))$ partitions. Consequently,

$$\begin{aligned} f(d;T) &\le \binom{n}{T}2^T d^{3T/4}\exp\Big((n-T)(\log d - 1 + \gamma) + O(nd^{-1/3})\Big) \\ &= \exp\Big(n(\log d - 1 + \gamma) + O(nd^{-1/3})\Big)\binom{n}{T}\Big(2e^{1-\gamma}d^{-1/4}\Big)^T, \end{aligned}$$

as required. □

We now prove Theorem 1.2.

*Proof of Theorem 1.2.* By Lemma 5.1 and the binomial theorem,

$$f(d) = \sum_{T=0}^{n} f(d;T) \le \exp\left(n(\log d - 1 + \gamma) + O(nd^{-1/3})\right)\left(1 + 2e^{1-\gamma}d^{-1/4}\right)^n. \tag{5.2}$$

Since $\log(1+x) \le x$ for $x \ge 0$, we obtain $\log f(d) \le n(\log d - 1 + \gamma) + O(nd^{-1/4})$. On the other hand, the lower bound established in Section 4 gives $\log R_d \ge n(\log d - 1 + \gamma) - O\left(nd^{-1/3}(\log d)^{4/3}\right)$. Moreover, taking $A = V(Q_d)$ in Proposition 3.3 yields $\log Z_d \le n(\log d - 1 + \gamma) + O(nd^{-1/3})$. Together with $R_d \le Z_d \le f(d)$, these estimates prove Theorem 1.2. □

## 6. The structure of a typical partition

In this section, we prove Theorem 1.4, which describes the typical structure of a uniformly random subcube partition. We show that the numbers of squares and edges concentrate around $\alpha n$ and $pn$, respectively, while singleton vertices and vertices covered by subcubes of dimension at least three form a vanishing proportion of $V(Q_d)$. The proof combines the counting estimates from Section 3 with a bound on the contribution of higher-dimensional subcubes, see Lemma 6.1.

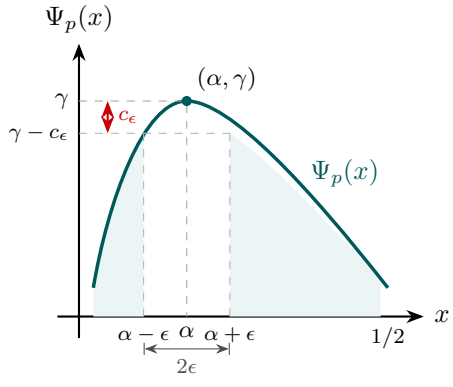
**Lemma 6.1.** *There is an absolute constant $K > 0$ such that, for $T_0 = \lceil Knd^{-1/4} \rceil$,*

$$\frac{1}{f(d)} \sum_{T \ge T_0} f(d;T) \le \exp(-\Omega(nd^{-1/4})). \tag{6.1}$$

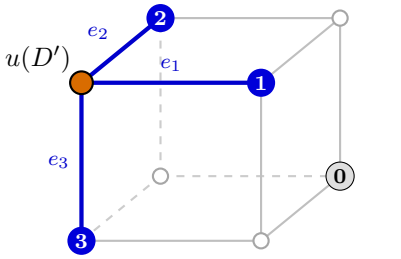
*Proof.* Choose $K > 4e^{2-\gamma}$. Using the bound $\binom{n}{T} \le (en/T)^T$, every $T \ge T_0$ satisfies

$$\binom{n}{T}(2e^{1-\gamma}d^{-1/4})^T \le 2^{-T}.$$

Summing (5.1) over all $T \ge T_0$ and dividing by the lower bound on $R_d \le f(d)$ gives (6.1), as both $nd^{-1/3}$ and $nd^{-1/3}(\log d)^{4/3}$ are $o(nd^{-1/4})$. □




**(a)** Strictly concave profile $\Psi_p(x)$




**(b)** Coordinate encoding on $Q_3$

Figure 4: Geometric intuition for Theorem 1.4. (a) A schematic illustration of the strictly concave function $\Psi_p(x)$, maximized at $\alpha$. (b) Coordinate encoding on a three-dimensional subcube.

Now we prove Theorem 1.4.

*Proof of Theorem 1.4.* Fix $\varepsilon > 0$. For a uniformly random partition into edges and squares, sum (3.4) with $A = V(Q_d)$ over all values satisfying $|s - \alpha n| \geq \varepsilon n$. By the lower bound in Theorem 1.2,

$$\mathbb{P}(|S_d - \alpha n| \geq \varepsilon n) \leq \exp(-\Omega_\varepsilon(n)). \tag{6.2}$$

Singletons and higher dimensional parts are completely absent in this family.

Next consider the family $\mathcal{P}_{\{0,1,2\}}(d)$ counted by $Z_d$. Let $t$ be the number of singleton vertices of each parity. By Lemma 3.1, $Z_{d,t} \leq \binom{n}{t} d^{n-t}$. For $x = (t/n) \geq (2/\log d)$, the function $h(x) - x \log d$ is strictly decreasing for large $d$, and its value at $(2/\log d)$ is $(-2 + o(1))$. Thus, using the lower bound from Theorem 1.2,

$$\frac{1}{Z_d} \sum_{t \geq 2n/\log d} Z_{d,t} \leq e^{-n/2} \tag{6.3}$$

for all sufficiently large $d$.

For $t < (2n/\log d)$, the set remaining after deleting the singletons is balanced with $b = n - t$ vertices of each parity. If $|s - \alpha n| \geq \varepsilon n$, then $|s - \alpha b| \geq \varepsilon n/2$ for large $d$. By (3.4), each such residual set has at most $\exp(n(\log d - 1 + \gamma) - \varepsilon^2 n/4 + o(n))$ partitions with any one such $s$. This error is uniform in the set, $t$, and $s$. The number of possible singleton sets in this range is bounded by

$$\sum_{t < 2n/\log d} \binom{n}{t}^2 = \exp\left(O\left(n \frac{\log\log d}{\log d}\right)\right) = \exp(o(n)).$$

There are at most $n + 1$ possible values for $s$. Together with (6.3) and the lower bound on $Z_d$, this proves that for some strictly positive constant $\eta = \eta(\varepsilon) > 0$, the number of partitions counted by $Z_d$ with $|S_d - \alpha n| \geq \varepsilon n$ is at most $Z_d e^{-\eta n}$. Equation (6.3) also implies that $U_d/n \to 0$ in probability in this family.

Finally, consider unrestricted partitions in $\mathcal{P}(d)$. By Lemma 6.1, we may lose an exceptional set of probability tending to zero and assume that $T = W_d/2 < T_0$. Replacing each higher dimensional part by its constituent singleton vertices defines a map to $\mathcal{P}_{\{0,1,2\}}(d)$ that preserves the number of squares and cannot decrease the number of singletons. For every image partition, the number of preimages with $T < T_0$ satisfies

$$\sum_{0 \leq T < T_0} \binom{n}{T} 2^T d^{3T/4} = \exp(O(n d^{-1/4} \log d)) = \exp(o(n)), \tag{6.4}$$

owing to the encoding established in Lemma 5.1. The exponentially small exceptional sets just established for $Z_d$ therefore produce only $o(f(d))$ preimages, because $f(d) \geq Z_d$. This establishes square concentration and the limit $U_d/n \to 0$ in probability for unrestricted partitions. The bound $W_d/n \to 0$ in probability follows directly from (6.1).

In all three partition families, the global vertex conservation identity

$$2E_d + 4S_d + U_d + W_d = 2n$$

now ensures that $E_d/n \to 1 - 2\alpha = p$ in probability, which completes the proof. □

## 7. Partitions into subcubes of a fixed dimension

We prove Theorem 1.5 by combining the estimate of Alon, Balogh, and Potapov [1, Proposition 1.7] with Lemma 2.3. The latter allows us to eliminate singleton parts while preserving the leading term in the logarithm of the count.

*Proof of Theorem 1.5.* Fix an integer $r \geq 2$. By Alon, Balogh, and Potapov [1, Proposition 1.7], we have $\log f_{0,r}(d) = (r/2^{r-1} + o(1))n \log d$. Since $f_r(d) \leq f_{0,r}(d)$, this proves the required upper bound.

For the lower bound, let $k = d - r$ and apply Lemma 2.3 with $\mathcal{A} = \mathcal{P}_{\{0,r\}}(k)$ and $\mathcal{D} = \{r\}$. Since $Q_k$ has $2^k$ vertices, there are at most $2^{2^k}$ possible singleton sets. Hence

$$f_r(d) \geq \frac{f_{0,r}(k)^{2^r}}{2^{(2^r-1)2^k}}. \tag{7.1}$$

The same result of Alon, Balogh, and Potapov implies that $\log f_{0,r}(k) = (r/2^{r-1} + o(1))2^{k-1} \log k$. Taking logarithms in (7.1), and noting that $2^r 2^{k-1} = n$ and $\log k \sim \log d$, we obtain $\log f_r(d) \geq (r/2^{r-1} + o(1))n \log d$, as required. □

For $r = 2$, the construction uses four parallel copies of $Q_k$. In each layer, we choose a partition with the same singleton set $S$, and for every $v \in S$ the corresponding singletons form the square $\{v\} \times Q_2$. The partitions in different layers need not coincide. Requiring their singleton sets to agree incurs a loss of only $O(2^k)$ in the logarithm of the count, which is negligible compared with $2^k \log k$.

A more precise upper bound follows from Lemma A.1. Let $\mathcal{H}$ be the hypergraph on $V(Q_d)$ whose edges are the vertex sets of the $r$-dimensional subcubes. Its perfect matchings correspond to the partitions counted by $f_r(d)$. This hypergraph is $2^r$-uniform and $\binom{d}{r}$-regular. Two vertices at Hamming distance $j \leq r$ belong to $\binom{d-j}{r-j}$ common edges, so its maximum codegree is $\binom{d-1}{r-1}$. Since $\binom{d-1}{r-1}/\binom{d}{r} = r/d$, Lemma A.1 yields

$$\log f_r(d) \leq \frac{2^d}{2^r}\left(\log\binom{d}{r} - (2^r - 1)\right) + O_r\left(2^d d^{-1/(2^r-1)}\right). \tag{7.2}$$

In particular, for $r = 2$ we obtain

$$\log f_2(d) \leq n \log d - \frac{3 + \log 2}{2} n + O(nd^{-1/3}). \tag{7.3}$$

The lower-bound argument above determines the term of order $n \log d$, but does not determine the coefficient of $n$ for pure square tilings.

## Appendix A. An entropy bound for labeled hypergraph matchings

We give a quantitative version of Luria's entropy argument [25, Theorem 3.1], allowing nonregular hypergraphs and distinguishable parallel edges. Related entropy methods appear in [28, 24].

**Lemma A.1.** *Fix an integer $r \geq 3$. Let $\mathcal{H}$ be an $r$ uniform multi hypergraph on $N$ vertices, with distinguishable parallel edges, maximum degree at most $K$, and maximum codegree $\Delta_2$, both counted with multiplicities. If the ratio satisfies*

$$\delta = \frac{1 + \binom{r}{2}\Delta_2}{K} \leq \frac{1}{2},$$

*then its number $M(\mathcal{H})$ of labeled perfect matchings satisfies*

$$M(\mathcal{H}) \leq \left(\frac{K}{e^{r-1}}\right)^{N/r} \exp(O_r(N\delta^{1/(r-1)})). \tag{A.1}$$

*Proof.* When $M(\mathcal{H}) = 0$, the inequality holds trivially. Otherwise, let $X$ be a uniformly chosen labeled perfect matching, and let $X_v$ denote its edge containing vertex $v$. Independently assign each vertex a uniform priority $U_v \in [0, 1]$, and expose the random variables $X_v$ in decreasing order of priority. With probability one, all vertex priorities are distinct.

For fixed priorities, let $\mathcal{F}_v$ represent the previously exposed information. The condition whether $v$ is already covered by a revealed matching edge is completely determined by $\mathcal{F}_v$. If vertex $v$ is already covered, the conditional entropy of $X_v$ vanishes. Otherwise, let $\mathcal{N}_v$ count the labeled edges through $v$ that are disjoint from all previously revealed matching edges. These candidates contain all possible values of $X_v$. Writing $A_v$ for the event that $v$ remains uncovered, the chain rule combined with the logarithmic bound on the support size yields

$$H(X) \leq \sum_v \mathbb{E}_{X,U}\big[\mathbf{1}_{A_v} \log \mathcal{N}_v\big]. \tag{A.2}$$

The event $A_v$ occurs precisely when $v$ possesses the largest priority in $X_v$.

Now condition on $X$ and the event $U_v = x$. The probability of $A_v$ equals $x^{r-1}$. Applying Jensen's inequality to the remaining priorities in (A.2) produces

$$H(X) \leq \mathbb{E}_X \sum_v \int_0^1 x^{r-1} \log \mathbb{E}_U\big[\mathcal{N}_v \mid X, U_v = x, A_v\big]\, dx. \tag{A.3}$$

This step conditions solely on the support size estimate, without replacing the conditional entropy by one conditioned on the entire matching.

For a fixed matching $X$, call a labeled hyperedge exceptional if it contains two vertices in a single edge of $X$. Let $B_v$ count the exceptional labeled edges incident to $v$. Since $X$ contains exactly $N/r$ edges, the total number of exceptional labeled hyperedges is at most $(N/r)\binom{r}{2}\Delta_2$. Consequently,

$$\sum_v B_v \leq N \binom{r}{2} \Delta_2. \tag{A.4}$$

A non exceptional edge through $v$ intersects $r-1$ distinct other edges of $X$, each in exactly one vertex. Such an edge remains available only if every vertex in those $r-1$ matching edges has priority strictly below $x$. Otherwise, at least one of those vertices would have revealed its matching edge earlier, excluding the candidate. Because they lie strictly outside $X_v$, the $r(r-1)$ priorities involved are mutually independent, even after conditioning on the event $A_v$. Therefore,

$$\mathbb{E}_U\big[\mathcal{N}_v \mid X, U_v = x, A_v\big] \leq 1 + B_v + K x^{r(r-1)}.$$

The additional unit provides a convenient positive lower bound throughout. Applying the concavity of the logarithm across all vertices $v$, together with the sum bound (A.4), we obtain

$$H(X) \leq N \int_0^1 x^{r-1} \log\big(K x^{r(r-1)} + K\delta\big)\, dx.$$

Factoring out $K$ reduces this integral to

$$H(X) \leq \frac{N}{r}\big(\log K - (r-1)\big) + N I_r(\delta),$$

where the remainder integral is given by

$$I_r(\delta) = \int_0^1 x^{r-1} \log\big(1 + \delta x^{-r(r-1)}\big)\, dx.$$

Applying the change of variables $x = \delta^{1/(r(r-1))} y$ transforms this remainder into

$$I_r(\delta) = \delta^{1/(r-1)} \int_0^{\delta^{-1/(r(r-1))}} y^{r-1} \log\big(1 + y^{-r(r-1)}\big)\, dy. \tag{A.5}$$

The integrand behaves as $O_r(y^{r-1}(1+|\log y|))$ near the origin and is bounded by $y^{-(r-1)^2}$ for all $y \geq 1$. Because $r \geq 3$, the integral over the entire half line $(0,\infty)$ converges to a finite constant depending only on $r$. Hence $I_r(\delta) = O_r(\delta^{1/(r-1)})$. Since $H(X) = \log M(\mathcal{H})$, the estimate (A.1) follows, which completes the proof. □

## Appendix B. Further counting consequences

We prove Corollary 1.6 and the fixed-coverage estimate stated in the introduction. We use [4, Theorem 2] in the following form. Fix an integer $h \geq 2$ and $0 < \varepsilon < 1/2$. If an $h$-uniform hypergraph on $N$ vertices has all degrees $(1+o(1))D$, where $D \to \infty$, and maximum codegree $o(D)$, then, for sufficiently large $D$, it has at least $\exp((1-2\varepsilon)(N/h)\log D)$ matchings, each of size at least $(1-\varepsilon)N/h$. In particular, its total number of matchings is at least $\exp((1-o(1))(N/h)\log D)$. All applications below are to exactly regular hypergraphs.

*Proof of Corollary 1.6.* Write $N = q^d$ and $B = ed^\beta$. For $A \subseteq [q]^d$, let $g(A)$ count partitions of $A$ with dimensions in $\mathcal{D}$. We first show by induction on $|A|$ that $g(A) \leq B^{|A|}$. The empty set has one partition. For nonempty $A$, fix a vertex $v$ and classify each partition by the part containing $v$. At most $\binom{d}{r}$ coordinate $r$-faces contain $v$. Hence the induction step follows from

$$\sum_{\substack{r\in\mathcal{D}\\ r\leq d}} \binom{d}{r} B^{-q^r} \leq \sum_{r\geq 0} \frac{e^{-q^r}}{r!} \leq e^{-1} \sum_{r\geq 0} \frac{1}{r!} = 1,$$

where $r - \beta q^r \leq 0$. Thus $\log f_{\mathcal{D}}^{(q)}(d) \leq \beta N \log d + N$.

Choose a fixed $r_* \in \mathcal{D}$ attaining $\beta$. Such an integer exists because $\beta > 0$ and $r/q^r \to 0$. On $[q]^k$, let $\mathcal{H}_k$ be the hypergraph whose edges are all coordinate $r_*$-faces. It is $Q = q^{r_*}$-uniform, has degree $D_k = \binom{k}{r_*}$, and has maximum codegree $\binom{k-1}{r_*-1}$. Its relative maximum codegree is $r_*/k \to 0$. By [4, Theorem 2], there is a family $\mathcal{A}_k$ of matchings with $\log|\mathcal{A}_k| \geq (\beta - o(1))q^k \log k$.

Group these matchings by their uncovered vertex sets. There are at most $K = 2^{q^k}$ groups. In the $Q$ parallel copies of $[q]^k$ in $[q]^k \times [q]^{r_*}$, choose matchings with a common uncovered set $U$. Cover the remaining vertices by the faces $\{u\} \times [q]^{r_*}$, $u \in U$. The resulting partition is purely $r_*$-dimensional. The construction is injective: new parts have all their free coordinates among the new coordinates, whereas old parts have them among the original coordinates. The power-mean inequality therefore yields $f_{\{r_*\}}^{(q)}(k+r_*) \geq |\mathcal{A}_k|^Q / K^{Q-1}$. The logarithmic loss is $O_{q,r_*}(q^{k+r_*})$, proving the lower bound after writing $d = k + r_*$.

It remains to prove the assertion about typical dimensions. If $\mathcal{D}\setminus I$ is empty, this is immediate. Otherwise define $g = \beta - \sup_{r\in\mathcal{D}\setminus I} r/q^r > 0$ and $\theta = g/2$. The gap is positive because $r/q^r \to 0$ and $\beta > 0$. For a partition $P$ of any $A$, let $W(P)$ be its number of vertices in parts of dimension outside $I$. The same induction gives $\sum_P d^{\theta W(P)} \leq B^{|A|}$: the normalized induction sum is at most $\sum_{r\in\mathcal{D}, r\leq d} \binom{d}{r} d^{\theta q^r \mathbf{1}_{\{r\notin I\}}} B^{-q^r} \leq 1$, using $r - (\beta-\theta)q^r \leq 0$ for dimensions outside $I$. Consequently, for each fixed $\varepsilon > 0$, $\mathbb{P}(W(P) \geq \varepsilon N) \leq \exp(-\theta\varepsilon N \log d + o(N\log d))$, where we divide by the first-order lower bound already proved. This completes the proof. □

*Proof of the fixed-coverage estimate.* Fix $q \geq 2$ and $0 < \theta < 1$. Write $N = q^d$, $m = \lfloor \theta N/q \rfloor$, and $L = Nd/q$, the total number of coordinate lines. The upper bound $K_{q,\theta}(d) \leq \binom{L}{m}$ implies $\log K_{q,\theta}(d) \leq (\theta/q)N\log d + O_{q,\theta}(N)$.

For the lower bound, fix $0 < \delta < (1-\theta)/2$. The line hypergraph is $q$-uniform and $d$-regular, and its maximum codegree is one. By [4, Theorem 2], it has a family $\mathcal{M}$ of at least $\exp((1-2\delta)(N/q)\log d)$ matchings, each containing at least $(1-\delta)N/q$ lines. From each member of $\mathcal{M}$, retain exactly $m$ lines according to a fixed ordering. Given the retained matching, each original matching is determined by at most $\lfloor N/q \rfloor - m$ additional lines. Hence each image has at most $\sum_{j=0}^{\lfloor N/q\rfloor - m} \binom{L}{j}$ preimages. The logarithm of this sum is at most $((1-\theta)/q)N\log d + O_{q,\theta}(N)$.

Thus $\log K_{q,\theta}(d) \geq ((\theta - 2\delta)/q)N \log d - O_{q,\theta}(N)$. Letting $d \to \infty$ and then $\delta \to 0$ proves the claim. □

## Declaration of generative AI

During the preparation and revision of this manuscript, OpenAI's ChatGPT was consulted to stress-test technical derivations, bridge intermediate arguments, and tedious calculation. Specifically, this assistance helped the authors detect an oversight in the parity-reduction step of Proposition 3.2, establish the uniform contraction estimate for the nonlinear operator in Lemma 4.2, and track the quantitative error dependence on the codegree parameter in Lemma A.1. In addition, ChatGPT was used to help write Python scripts for finite computational checks, suggest stylistic refinements to improve readability, and assist with preliminary reference checking. The authors reviewed and verified the mathematical arguments and references and take full responsibility for the final content.

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